

CAPSTONE PROJECT REGISTER

Class: Duration time: from/01/2025 To/06/2025

(*) Profession: Software Engineer Specialty: <JS> <IS> <SE>

(*) Kinds of person make registers: Lecturer Students

1. Register information for supervisor (if have)

No.	Fullname	Phone	E-Mail	Title
1	[Tên GVHD 1]	[SĐT]	[email@fe.edu.vn]	Supervisor
2	[Tên GVHD 2]	[SĐT]	[email@fe.edu.vn]	Supervisor

2. Register information for students (if have)

No.	Full name	Student code	Phone	E-mail	Role
1	[Họ tên Leader]	[MSSV]	[SĐT]	[email@fpt.edu.vn]	Leader
2	[Họ tên 2]	[MSSV]	[SĐT]	[email@fpt.edu.vn]	Member
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3. Register content of Capstone Project

3.1. Capstone Project name:

English: B2B Travel Platform — End-to-End Booking & Delivery

Vietnamese: Nền tảng B2B Du lịch — Đặt chỗ và Bàn giao Dịch vụ Trọn gói

Abbreviation: B2BTP (B2B Travel Platform)

Context:

- The travel distribution market in Vietnam remains heavily fragmented, with over 70% of transactions still processed through small and medium-sized travel agencies (sub-agents / freelance agents) who rely on manual, phone-based communication with suppliers for booking and pricing — spending 30 minutes to 2 hours per quote.
- Traditional travel agents face critical operational pain points: risk of losing held seats due to payment delays, error-prone manual markup calculations, and loosely managed paper/PDF vouchers vulnerable to fraud, forgery, and double-use at points of service delivery.
- There is currently no comprehensive B2B SaaS platform in Vietnam integrating the full booking workflow — from product search (hotels, flights, tours, bus tickets), inventory hold, agency wallet payment, digital voucher issuance, to end-to-end service delivery verification — within a single unified system.
- Existing Global Distribution Systems (GDS: Amadeus, Sabre) require prohibitively high deposit requirements, making them inaccessible for small travel agencies. Meanwhile, B2C OTA platforms (Agoda, Booking.com) do not support net-rate wholesale pricing or agent-to-agent commission structures.

- The absence of a secure digital voucher system means tour operators and transportation providers cannot reliably verify service authenticity at the point of delivery, creating opportunities for voucher forgery and duplicate usage that result in significant revenue leakage.
- As the Vietnamese travel market grows post-pandemic, there is a strong opportunity for a technology platform that lowers the barrier to professional travel distribution for small agencies while providing robust financial controls, real-time inventory management, and cryptographically secured service delivery verification.

Proposed Solutions:

Building a comprehensive B2B Travel Platform using a SaaS architecture, combining a web application for travel agents and suppliers, a mobile application for delivery drivers, and a secure QR-based E-Voucher system with the following feature groups:

Travel Agent (Web Application)

- Multi-product search: real-time availability and net pricing for hotels, flights, tours, and bus tickets.
- Inventory hold (Hold Booking) with automated release timer (15-minute countdown) and SignalR real-time status updates.
- Agency digital wallet (E-Wallet) with top-up via VNPay, pessimistic locking to prevent double-spending on concurrent sessions.
- Automatic price Markup configuration (percentage or fixed amount per product category), with Price Freeze enforced at booking creation.
- Combo booking with Saga Orchestrator pattern (Tour-First strategy): supplier approval before GDS PNR hold to prevent PNR timeout conflicts.
- E-Voucher issuance with HMAC-SHA256 cryptographic signing (Per-driver Key, 24-hour rotation) and dynamic QR generation (refreshes every 30 seconds).
- Booking history, wallet statement, financial reports, and CTV (collaborator) commission management.

Supplier (Web Portal)

- Product listing and real-time inventory management (available slots, pricing, service dates).
- On-Request booking approval workflow with Dynamic Cut-off Time: $\text{MIN}(\text{request_time} + 3 \text{ hours}, 21:00 \text{ the night before departure})$.
- Multi-branch service management with location-specific configuration.
- Revenue reporting, booking pipeline analytics, and occupancy trend dashboard.

Delivery Driver (Mobile Application — iOS & Android)

- Daily voucher synchronization at start of shift (requires internet connection); Per-driver HMAC-SHA256 key downloaded and stored AES-256 encrypted in Hive local database.
- Offline QR voucher scanning and cryptographic verification using locally stored driver key — no internet required at point of service.
- Dynamic QR timestamp validation: automatically rejects vouchers with timestamp drift > 5 minutes to prevent screenshot/screenshot-sharing fraud.
- GPS-tagged check-in logging with automatic background sync to server within 60 seconds of connectivity restoration.
- Conflict resolution on sync: 'First-write-wins' rule; duplicate check-in attempts escalated to fraud_alerts table for Admin review.

Platform Admin (Web Application)

- Agency and supplier account management with KYC document verification; periodic re-verification every 12 months (suspended on expiry).
- Financial audit dashboard: Double-entry Ledger enforcement, Real-time Balance Assertion after every transaction.

- Nightly Reconciliation Engine (00:05): batch comparison of ledger sum vs. cached balance for all agencies; Telegram alert on discrepancy.
 - Rate Limiting (1,000 req/min per agency), Audit Logs, and compliance reporting.
 - Per-driver Key revocation: immediate invalidation of all QR vouchers on a compromised device.
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Functional requirement

Travel Agent

(Web Application — authenticated session with JWT)

MainFlow:

- Search and compare travel services (hotel, flight, tour, bus) with real-time availability and net pricing.
- Hold booking with 15-minute countdown timer and automatic inventory release on expiry via Delayed Job.
- Top up agency wallet via VNPay QR; track wallet balance and transaction history.
- Confirm booking and pay with agency wallet (pessimistic locking prevents double-spending).
- Configure price Markup (percentage or fixed) per product and per customer segment.
- Book combo services (flight + tour) via Saga Orchestrator workflow (Tour-First strategy).
- Receive digitally signed E-Vouchers; forward to end customers via QR code.
- Request booking cancellation; track refund status (GDS cancellation + wallet credit).
- View booking history, financial statements, and generate CSV/Excel reports.
- Manage CTV (collaborator) sub-accounts and commission payout structures.

Supplier

(Web Portal — authenticated session with JWT)

MainFlow:

- Register and manage service listings (tours, transportation, accommodation).
- Set pricing, inventory slots, and departure/availability dates.
- Receive On-Request booking requests and approve/reject within the cut-off window.
- View booking pipeline, revenue analytics, and occupancy utilization reports.
- Manage multi-branch and location-specific service configurations.

Delivery Driver

(Mobile Application — authenticated with Per-driver Key)

MainFlow:

- Log in and synchronize the day's assigned vouchers and encrypted HMAC key (requires internet at start of shift).
- Scan customer QR vouchers offline; verify authenticity via local HMAC-SHA256 computation.
- Validate QR timestamp freshness; reject vouchers with drift > 5 minutes.
- Record GPS coordinates and exact timestamp for each successful check-in.
- Auto-sync offline check-in logs to server when internet connection is restored (within 60 seconds).
- View daily assignment schedule and per-booking delivery status.

Platform Admin

(Web Application — system administrator access)

MainFlow:

- Onboard and approve agency/supplier accounts with KYC document review.
- Suspend or revoke accounts upon KYC expiry, policy violations, or fraud detection.
- Monitor financial integrity: Double-entry Ledger dashboard, Real-time Balance Assertion alerts.
- Review nightly Reconciliation Engine reports; manually resolve flagged discrepancies.
- Configure Rate Limiting thresholds; review immutable Audit Logs.
- Revoke compromised Per-driver Keys; review fraud_alerts queue (duplicate check-ins, abnormal booking patterns).

- Manage system-wide settings: Cut-off Time policies, Markup constraints, KYC re-verification schedules.
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Non-functional requirement:

- Performance: Booking search API must return results within 2 seconds for 95% of requests under normal load. Wallet debit/credit must complete within 500ms with lock acquisition.
 - Scalability: System must support 500+ concurrent travel agent sessions without performance degradation, using horizontal API scaling and Redis caching layer.
 - Availability: 99.9% uptime SLA annually. Critical financial modules (wallet, ledger) must implement circuit breakers for graceful degradation under external service failures.
 - Security: HTTPS (TLS 1.3) for all APIs. JWT tokens expire in 24 hours with Refresh Token support. Per-driver QR keys rotate every 24 hours; stored AES-256 encrypted on devices. Rate Limiting: 1,000 req/min per authenticated agency to prevent abuse.
 - Financial Integrity: Double-entry Ledger for all wallet operations. Real-time Balance Assertion after every transaction (immediate wallet lock on discrepancy). Nightly Reconciliation Engine at 00:05 with Telegram Finance Team alert.
 - Offline Resilience: Driver mobile app must support full offline QR verification and GPS-tagged logging. Automatic sync to server within 60 seconds of connectivity restoration.
 - Data Consistency: Saga Orchestrator must implement compensating transactions for all combo booking failures. Price Freeze enforced at booking creation — prices cannot retroactively change for confirmed bookings.
 - Monitoring & Auditability: Structured logging (Serilog), distributed tracing, and immutable Audit Logs for all admin actions.
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3.2. Main proposal content (including result and product)

Theory and practice (document):

Students must apply the software development process, UML 2.0, and distributed systems design principles (Saga Pattern, State Machine, HMAC-SHA256 security). Required documents: User Requirements, SRS, Architecture Design, Detail Design, System Implementation, Testing Document, Installation Guide, source code, and deployable software packages.

Server-side technologies:

- Backend API: C# ASP.NET Core (Web API).
- Database: PostgreSQL (relational data, Double-entry Ledger, pessimistic locking with SELECT FOR UPDATE).
- Caching & Messaging: Redis (session cache, Pub/Sub real-time events, Distributed Lock for concurrent booking).
- Background Jobs: Hangfire (15-minute Hold expiry, Saga Orchestrator steps, nightly Reconciliation Engine at 00:05).
- Real-time Communication: SignalR WebSocket (booking status updates, inventory change notifications).
- Payment Gateway: VNPay IPN Webhook + Active Query Reconciliation (QueryDR API polling every 5 minutes for missed webhooks).
- Security: HMAC-SHA256 Per-driver Key signing, AES-256 local key encryption on mobile, JWT RS256.

Client-side technologies:

- Web Application (Agent, Supplier & Admin Portal): React.js / Next.js.
- Mobile Application (Delivery Driver App): Flutter (iOS & Android cross-platform).
- Offline Local Storage (Driver App): Hive Local NoSQL Database (AES-256 encrypted).

Products:

- Web Application — Travel Agent & Agency Manager Portal.
- Web Application — Supplier Management Portal.
- Web Application — Platform Admin Dashboard with Financial Audit.
- Mobile Application — Delivery Driver App (iOS & Android).

Proposed Tasks:

- Task 1: Backend API — Core booking engine, Inventory management, Agency Wallet service, Saga Orchestrator for combo bookings.
- Task 2: Web App for Travel Agents — Search, Hold Booking, Combo Booking, Markup configuration, E-Voucher issuance.
- Task 3: Web App for Suppliers — Product listing, On-Request approval workflow, Cut-off Time enforcement, revenue analytics.
- Task 4: Web App for Platform Admin — KYC management, Financial Audit Dashboard, Reconciliation Engine monitoring, Key revocation.
- Task 5: Mobile App for Delivery Drivers — Offline QR verification (HMAC-SHA256), GPS-tagged check-in, auto-sync, conflict resolution.
- Task 6: Security Layer — Per-driver Key management, 24-hour Key Rotation, AES-256 device encryption, Rate Limiting, Audit Logs.
- Task 7: Build – Deploy – Test — Docker containerization, CI/CD pipeline, load testing for concurrent booking and wallet scenarios.
- Task 8: Documentation — System Analysis & Design (State Machine, Saga Pattern, QR Security diagrams), Test Plan, Installation Manual, User Manual.

4. Other comments (propose all relative things if have).

The platform incorporates enterprise-grade financial controls (Double-entry Ledger, Real-time Balance Assertion, nightly Reconciliation Engine) and a cryptographically secured offline service delivery mechanism (Per-driver HMAC-SHA256 with 24-hour Key Rotation and AES-256 device encryption) typically found only in large-scale fintech and logistics systems. The Saga Orchestrator Tour-First design pattern resolves the PNR timeout conflict inherent in multi-service travel bookings. These decisions directly address real-world pain points of the Vietnamese travel distribution market and demonstrate advanced competency in distributed systems, financial integrity, and mobile security.

Supervisor (If have)	On behalf of Registers
	HCM, date /20 ...
(Sign and full name)	(Sign and full name)